

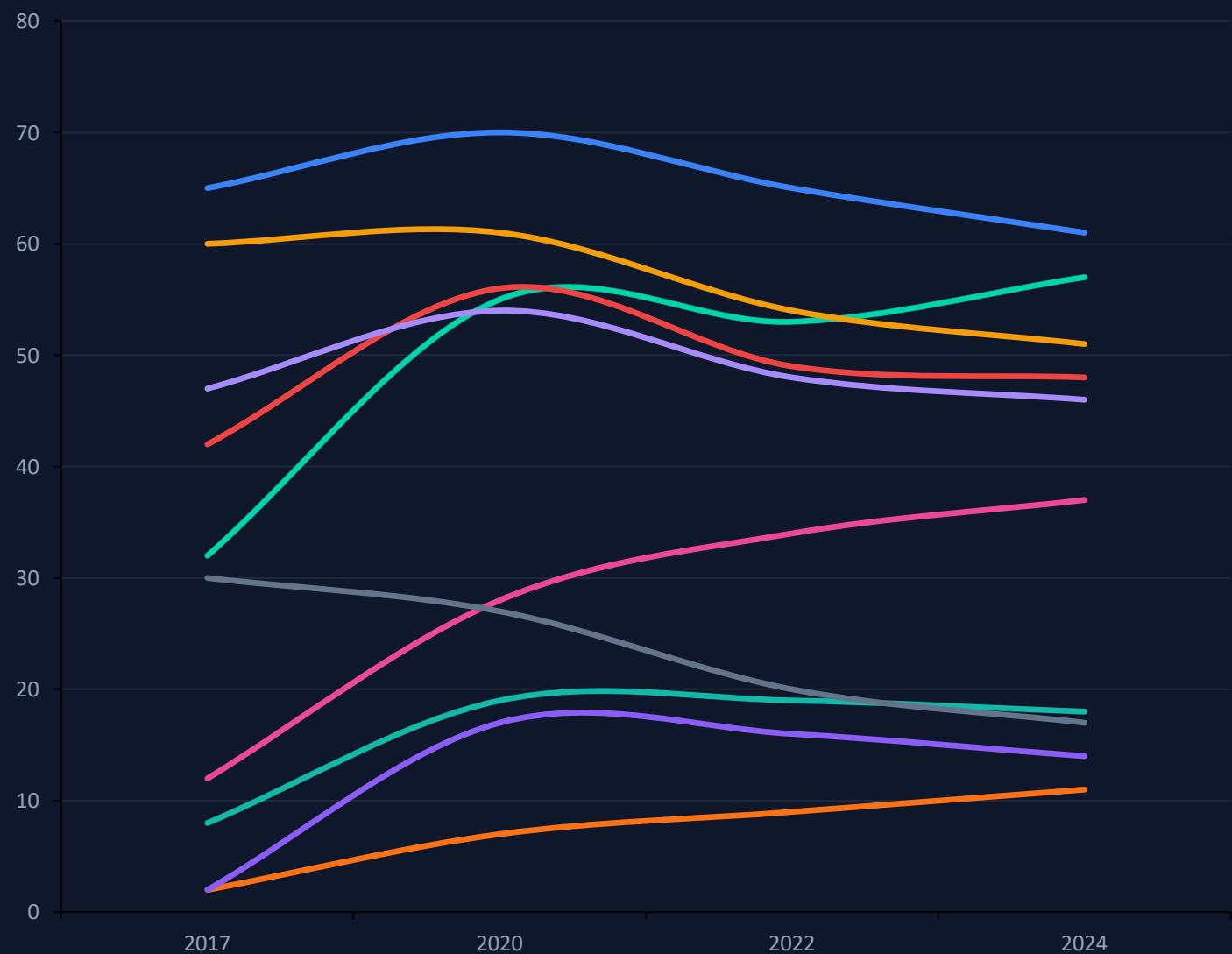
State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

Key insights from JetBrains' survey of 23,262 developers on programming language usage, satisfaction, adoption intent, and domain specialization trends from 2017 to 2024.

For Engineering Leaders & CTOs | April 2026

Python Closes the Gap: 8 Years of Language Usage Shifts



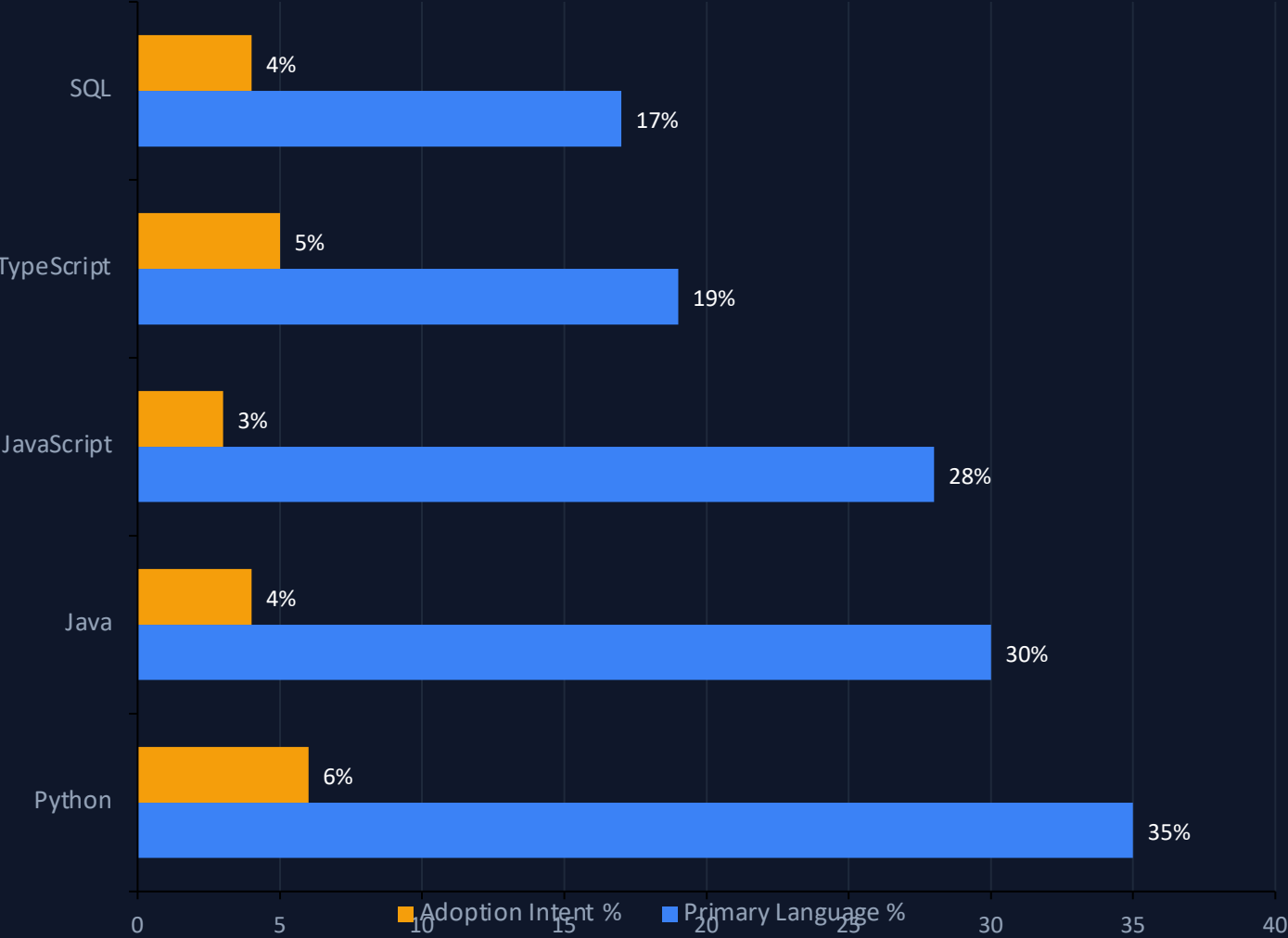
▲ Python 32% → 57%
Nearly closed the gap with JS

▲ TypeScript 12% → 37%
Tripled its share in 7 years

▼ PHP 30% → 17%
Significant long-term decline

Source: JetBrains State of Developer Ecosystem 2024 (n = 23,262)

Python Is Now the #1 Primary Language at 35%



Ecosystem Lock-In Effect

Adoption intent remains below 6% across all languages. Developers rarely plan to switch their primary stack — expertise investment and ecosystem familiarity create strong retention.

"Developers stick with what they know."
— JetBrains DevEco 2024

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Key Insight

Type-safe languages with modern tooling dominate the top spots. TypeScript and Rust both offer expressive type systems and strong developer experience.

Factor satisfaction into your retention strategy — devs stay where the tooling is great.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Leading Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs	Python	38%

Domain Drives Language Choice

- Python dominates AI/ML and data
- TypeScript/JS own the UI layer
- Java leads backend services
- C++ holds systems programming

Language choice is driven by domain requirements, not general preference.

What This Means for Your Technology Strategy in 2025

01

Invest in Python for AI/ML

Python is the undisputed leader in AI (31%), data science (28%), and data processing. Any AI/ML team investment starts here.

02

TypeScript is the future of UI

With 62% share in UI development and tripling usage since 2017, TypeScript is the strategic choice for frontend and full-stack.

03

Rust & Go for infrastructure

Both languages are the top growth plays for systems programming, cloud-native tooling, and microservices where performance is non-negotiable.

04

PHP & Ruby pools are shrinking

PHP dropped from 30% to 17%; Ruby from 10% to 4%. Plan for tightening talent pools in these ecosystems.

05

Satisfaction = Retention

TypeScript (199) and Rust (198) top satisfaction. Modern tooling and type safety correlate with developer happiness — factor this into retention.